Q1 - 24 January - Shift 1

Two long straight wires P and Q carrying equal current 10A each were kept parallel to each other at 5 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1. If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be :

(1) $8 F_1$ (2) $10 F_1$
(3) $F_1/8$ (4) $F_1/10$

Space for your notes:

Q2 - 24 January - Shift 1

A circular loop of radius r is carrying current I A. The ratio of magnetic field at the centre of circular loop and at a distance r from the center of the loop on its axis is :

(1) $1:3\sqrt{2}$ (2) $3\sqrt{2}:2$
(3) $2\sqrt{2}:1$ (4) $1:\sqrt{2}$

Space for your notes:

Q3 - 24 January - Shift 2

A long solenoid is formed by winding 70 turns cm^{-1}. If 2.0 A current flows, then the magnetic field produced inside the solenoid is ______
($\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1})

(1) 1232×10^{-4} T
(2) 176×10^{-4} T
(3) 352×10^{-4} T
(4) 88×10^{-4} T

Space for your notes:

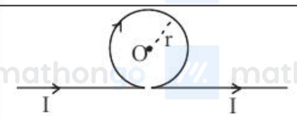
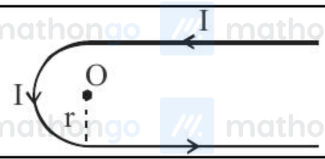
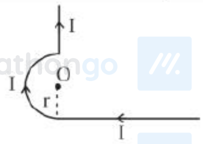
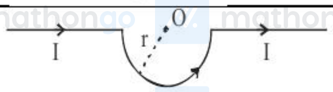
Q4 - 24 January - Shift 2

A single turn current loop in the shape of a right angle triangle with sides 5 cm, 12 cm, 13 cm is carrying a current of 2A. The loop is in a uniform magnetic field of magnitude 0.75 T whose direction is parallel to the current in the 13 cm side of the loop. The magnitude of the magnetic force on the 5 cm side will be $\frac{x}{130}$ N. The value of x is

Space for your notes:

Q5 - 25 January - Shift 1

Math List I with List II

List – I (Current configuration)		List – II (Magnetic field at point O)	
A		I.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
B		II.	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
C		III.	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
D		IV	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

Choose the correct answer from the option given below:

(1) A-III, B-IV, C-I, D-II

(2) A-I, B-III, C-IV, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-II, B-I, C-IV, D-III

Space for your notes:

Q6 - 25 January - Shift 1

A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2 m long and 0.2 m in diameter. The magnetic intensity at the center of the solenoid when a current of 2 A flows through it is:
(1) 2.4×10^{3} A m^{-1}
(2) 1.2×10^{3} A m^{-1}
(3) 1 A m^{-1}
(4) 2.4×10^{-3} A m^{-1}

Space for your notes:

Q7 - 25 January - Shift 2

For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10 mA is passed through it. If the torsional constant of suspension wire is 4.0×10^{-5} Nm rad^{-1}, the magnetic field is 0.01 T and the number of turns in the coil is 200, the area of each turn (in cm^2) is :
(1) 2.0 (2) 1.0
(3) 1.5 (4) 0.5

Space for your notes:

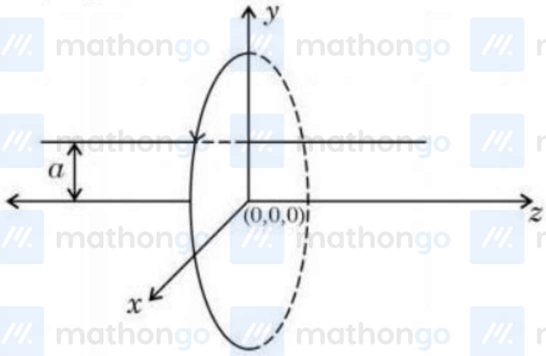
Q8 - 25 January - Shift 2

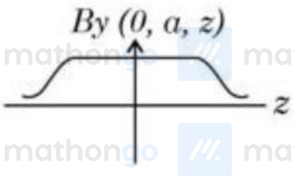
Two long parallel wires carrying currents 8A and 15 A in opposite directions are placed at a distance of 7 cm from each other. A point P is at equidistant from both the wires such that the lines joining the point P to the wires are perpendicular to each other. The magnitude of magnetic field at P is $\times 10^{-6}$ T. (Given : $\sqrt{2} = 1.4$)

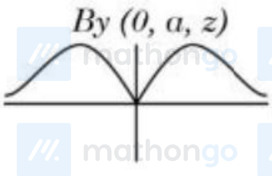
Space for your notes:

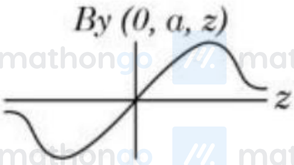
Q9 - 29 January - Shift 1

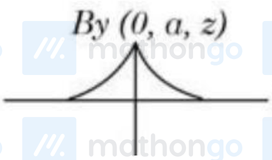
A single current carrying loop of wire carrying current I flowing in anticlockwise direction seen from +ve z direction and lying in xy plane in shown in figure. The plot of $\hat{j}$ component of magnetic field (By) at a distance 'a' (less than radius of the coil) and on yz plane vs z coordinate look like



(1) 

(2) 

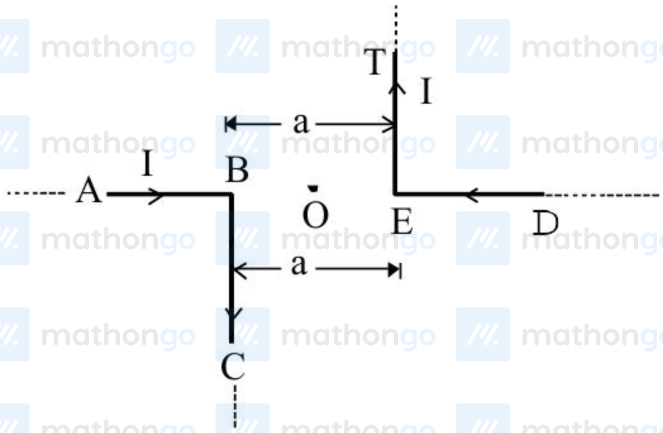
(3) 

(4) 

Space for your notes:

Q10 - 29 January - Shift 1

The magnitude of magnetic induction at mid-point O due to current arrangement as shown in Fig will be :



(1) $\frac{\mu_0 I}{2\pi a}$ (2) 0

(3) $\frac{\mu_0 I}{4\pi a}$ (4) $\frac{\mu_0 I}{\pi a}$

Space for your notes:

Q11 - 29 January - Shift 2

A square loop of area $25 cm^2$ has a resistance of 10Ω. The loop is placed in uniform magnetic field of magnitude 40.0 T. The plane of loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be

(1) 2.5×10^{-3} J (2) 1.0×10^{-3} J

(3) 1.0×10^{-4} J (4) 5×10^{-3} J

Space for your notes:

Q12 - 29 January - Shift 2

The electric current in a circular coil of four turns produces a magnetic induction 32 T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be :

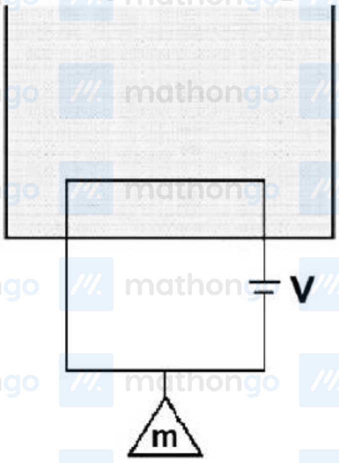
(1) 8T (2) 4T

(3) 2T (4) 16T

Space for your notes:

Q13 - 30 January - Shift 1

A massless square loop, of wire of resistance $10\,\Omega$, supporting a mass of 1 g, hangs vertically with one of its sides in a uniform magnetic field of 10^3 G, directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V, the magnetic force will exactly balance the weight of the supporting mass of 1g?
(If sides of the loop = 10 cm, g = 10 ms^{-2})



(1) $\frac{1}{10}$ V (2) 100 V

(3) 1 V (4) 10 V

Space for your notes:

Q14 - 30 January - Shift 1

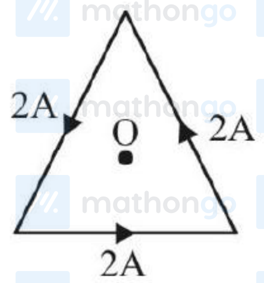
The magnetic moments associated with two closely wound circular coils A and B of radius $r_A = 10$ cm and $r_B = 20$ cm respectively are equal if: (Where N_A, I_A and N_B, I_B are number of turn and current of A and B respectively)

(1) $2N_AI_A = N_BI_B$ (2) $N_A = 2N_B$
(3) $N_AI_A = 4N_BI_B$ (4) $4N_AI_A = N_BI_B$

Space for your notes:

Q15 - 30 January - Shift 2

As shown in the figure, a current of 2A flowing in an equilateral triangle of side $4\sqrt{3}$ cm. The magnetic field at the centroid O of the triangle is :



(Neglect the effect of earth's magnetic field.)

(1) $4\sqrt{3}\times10^{-4}$ T (2) $4\sqrt{3}\times10^{-5}$ T
(3) $\sqrt{3}\times10^{-4}$ T (4) $3\sqrt{3}\times10^{-5}$ T

Space for your notes:

Q16 - 30 January - Shift 2

A current carrying rectangular loop PQRS is made of uniform wire. The length PR = QS = 5 cm and PQ = RS = 100 cm. If ammeter current reading changes from I to 2I, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $f_{PQ}^{I} : f_{PQ}^{2I}$ is :

(1) 1 : 2
(2) 1 : 4
(3) 1 : 5
(4) 1 : 3

Space for your notes:

Q17 - 31 January - Shift 1

A rod with circular cross-section area 2 cm^2 and length 40 cm is wound uniformly with 400 turns of an insulated wire. If a current of 0.4 A flows in the wire windings, the total magnetic flux produced inside windings is $4\pi\times10^{-6}$ Wb. The relative permeability of the rod is

(Given : Permeability of vacuum $\mu_0 = 4\pi\times10^{-7}\ NA^{-2}$)

(1) 12.5

(2) $\frac{32}{5}$

(3) 125

(4) $\frac{5}{16}$

Space for your notes:

Q18 - 31 January - Shift 1

A bar magnet with a magnetic moment 5.0 Am^2 is placed in parallel position relative to a magnetic field of 0.4 T. The amount of required work done in turning the magnet form parallel to antiparallel position relative to the field direction is ________.

(1) 4 J (2) 1 J

(3) 2 J (4) Zero

Space for your notes:

Q19 - 31 January - Shift 2

A long conducting wire having a current I flowing through it, is bent into a circular coil of N turns. Then it is bent into a circular coil of n turns. The magnetic field is calculated at the centre of coils in both the cases. The ratio of the magnetic field in first case to that of second case is:

(1) N : n (2) $n^2 : N^2$

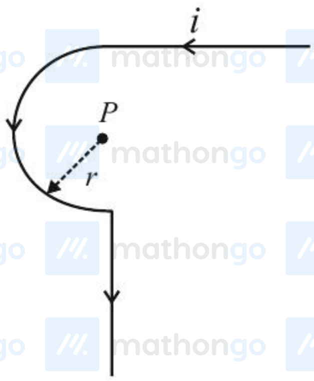
(3) $N^2 : n^2$ (4) n : N

Space for your notes:

Q20 - 01 February - Shift 1

Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to two long straight wires.

Space for your notes:



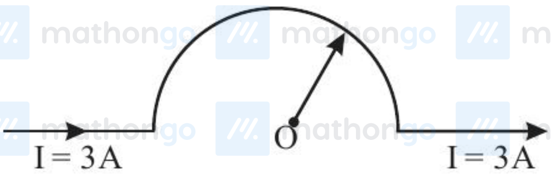
(1) $\frac{\mu_0 i}{2r}\left(1+\frac{2}{\pi}\right)$ (2) $\frac{\mu_0 i}{2r}\left(1+\frac{1}{\pi}\right)$

(3) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$ (4) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{\pi}\right)$

Q21 - 01 February - Shift 2

As shown in the figure, a long straight conductor with semicircular arc of radius $\frac{\pi}{10}$ m is carrying current I = 3A. The magnitude of the magnetic field. at the center O of the arc is:
(The permeability of the vacuum = $4\pi \times 10^{-7}$ NA^{-2})

Space for your notes:



(1) 6μT (2) 1μT

(3) 4μT (4) 3μT

Q22 - 01 February - Shift 2

A coil is placed in magnetic field such that plane of coil is perpendicular to the direction of magnetic field. The magnetic flux through a coil can be changed:

A. By changing the magnitude of the magnetic field within the coil.

B. By changing the area of coil within the magnetic field.

C. By changing the angle between the direction of magnetic field and the plane of the coil.

D. By reversing the magnetic field direction abruptly without changing its magnitude.

Choose the most appropriate answer from the options given below:

(1) **A** and **B** only
(2) **A**, **B** and **C** only
(3) **A**, **B** and **D** only
(4) **A** and **C** only

Space for your notes:

Q23 - 01 February - Shift 2

A square shaped coil of area 70 cm^2 having 600 turns rotates in a magnetic field of 0.4 wbm^{-2}, about an axis which is parallel to one of the side of the coil and perpendicular to the direction of field. If the coil completes 500 revolution in a minute, the instantaneous emf when the plane of the coil is inclined at 60° with the field, will be ____ V.

(Take $\pi = \frac{22}{7}$)

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (1)	**Q2** (3)	**Q3** (2)	**Q4** (9)
Q5 (3)	**Q6** (2)	**Q7** (2)	**Q8** (68)
Q9 (3)	**Q10** (4)	**Q11** (2)	**Q12** (3)
Q13 (4)	**Q14** (3)	**Q15** (4)	**Q16** (2)
Q17 (3)	**Q18** (1)	**Q19** (3)	**Q20** (3)
Q21 (4)	**Q22** (2)	**Q23** (44)	

Q1 (1)

Force per unit length between two parallel straight wires $= \frac{\mu_0 i_1 i_2}{2\pi d}$

$$\frac{F_1}{F_2} = \frac{\frac{\mu_0 (10)^2}{2\pi(5\text{cm})}}{\frac{\mu_0 (20)^2}{2\pi\left(\frac{5\text{cm}}{2}\right)}} = \frac{1}{8}$$

$\Rightarrow \ F_2 = 8F_1$

Q2 (3)

Magnetic field due to current carrying circular loop on its axis is given as

$$\frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}}$$

At centre, $x = 0, \ B_1 = \frac{\mu_0 i}{2r}$

At $x = r, \ B_2 = \frac{\mu_0 i}{2 \times 2\sqrt{2} r}$

$$\frac{B_1}{B_2} = 2\sqrt{2}$$

Q3 (2)

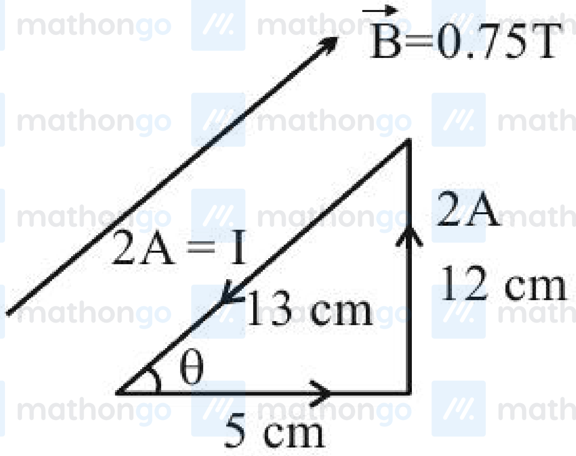
$$B = \mu_0 nI$$

$$= 4\pi \times 10^{-7} \times 70 \times 10^2 \times 2$$

$$= 56\pi \times 10^{-4}\ T$$

$$= 176 \times 10^{-4}\ T$$

Q4 (9)



Force on 5 cm side is

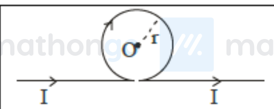
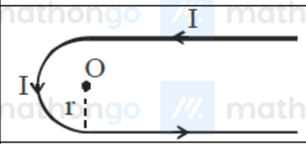
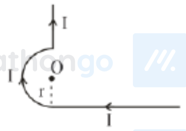
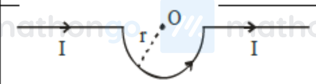
$$|\vec{F}| = ILB\sin\theta$$

$$= (2)(5 \times 10^{-2}) \times \frac{3}{4} \times \frac{12}{13} = \frac{9}{130} N$$

So, x = 9

Q5 (3)

Math List I with List II

List – I (Current configuration)		List – II (Magnetic field at point O)	
A		I.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
B		II.	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
C		III	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
D		IV	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

Choose the correct answer from the option given below:

(1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-I, C-IV, D-III

Q6 (2)

Magnetic field at centre inside the solenoid is given by

$B = \mu_0 nI$

So magnetic intensity at centre

$$H = \frac{B}{\mu_0} = nI = \left(\frac{1200}{2}\right)(2)$$

$H = 1.2 \times 10^3\ Am^{-1}$

Q7 (2)

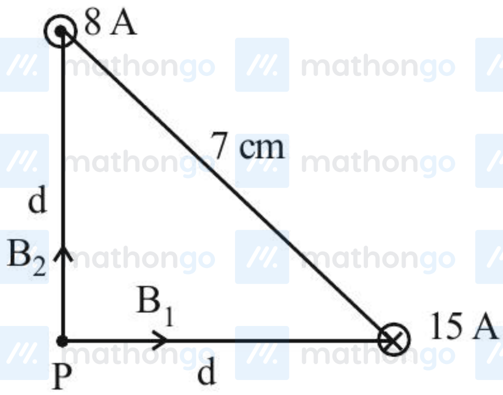
$\tau = K\theta$

$NiAB = K\theta$

$$A = \frac{K\theta}{NiB} = \frac{4\times10^{-5}\times0.05}{200\times10\times10^{-3}\times0.01}$$

On solving $A = 10^{-4}\ m^2 = 1\ cm^2$

Q8 (68)



Magnetic fields due to both wires will be perpendicular to each other.

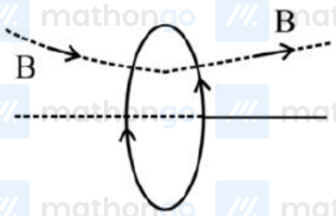
$B_1 = \frac{\mu_0 i_1}{2\pi d}$ $B_2 = \frac{\mu_0 i_2}{2\pi d}$

$B_{net} = \sqrt{B_1^2 + B_2^2} \Rightarrow \frac{\mu_0}{2\pi d}\sqrt{i_1^2 + i_2^2}$

$\Rightarrow \frac{4\pi\times10^{-7}}{2\pi\times(7/\sqrt{2})\times10^{-2}}\times\sqrt{8^2+15^2}$ $(d = \frac{7}{\sqrt{2}}$ cm)

$\Rightarrow 68\times10^{-6}$ T

Q9 (3)



$B_y = 0$ in plane of coil

B_y is opposite of each other in –z and +z positions.

Q10 (4)

Magnetic field due to current in BC and ET are outward at point 'O'

$$B_0 = \frac{\mu_0 i}{4\pi r} + \frac{\mu_0 i}{4\pi r} = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 i}{\pi a}$$

Q11 (2)

$\ell = 50$ cm

$t = 1$ sec

$$\therefore \ V = \frac{0.05}{1} = 0.05 \text{ m/s.}$$

$$i = \frac{40 \times 0.05 \times 0.05}{10} = 0.01\text{A}$$

$$F = B_i\ell = 40 \times 0.01 \times 0.05$$

$$F = 0.02 \text{ N}$$

$$\therefore \quad W = 0.02 \times \ell = 0.02 \times .05$$

$$\therefore \quad W = 1 \times 10^{-3}\text{J}$$

Q12 (3)

$$B = \frac{\mu_0 1}{2R} \times 4$$

$$B' = \frac{\mu_0 i}{2R'}$$

$$R' = 4R$$

$$B' = \frac{\mu_0 i}{8R}$$

$$\frac{B'}{B} = \frac{1}{16}$$

$$B' = 2T$$

Q13 (4)

$$F_m = mg$$

$$\therefore \quad ILB = mg$$

$$\therefore \quad \left(\frac{V}{R}\right) LB = mg$$

$$\therefore \quad V = \frac{mgR}{LB}$$

$$= \frac{(1 \times 10^{-3}\ kg)(10\ m/s^2)(10\Omega)}{(0.1\ m)(10^3 \times 10^{-4}\ T)} = 10\ V$$

Q14 (3)

$M = NIA$

$M_A = M_B$

$\therefore \quad N_A I_A A_A = N_B I_B A_B$

$\therefore \quad N_A I_A \ \pi(0.1)^2 = N_B I_B \pi(0.2)^2$

$\therefore \quad N_A I_A = 4 N_B I_B$

Q15 (4)

$$d \tan 60° = 2\sqrt{3}$$

$$d = 2 \text{ cm}$$

$$B = 3 \times \frac{\mu_0 i}{2\pi d} \sin 60°$$

$$= 3 \times \frac{2 \times 10^{-7} \times 2}{2 \times 10^{-2}} \times \frac{\sqrt{3}}{2}$$

$$= 3\sqrt{3} \times 10^{-5}$$

Q16 (2)

$$F \propto I_1 I_2$$

$$F_I : F_{2I} = 1 : 4$$

Q17 (3)

$$\phi = \mu_r \mu_o \frac{N}{\ell} I \times A$$

$$\mu_r = 125$$

Option 3.

Q18 (1)

$$u = -MB\cos\theta$$

$$W = \Delta u$$

$$W = -MB\cos 180^o (-mB\cos 0^o)$$

$$W = 2\,MB = 2 \times 5 \times 0.4 = 4J$$

Option 1

Q19 (3)

No Solution

Q20 (3)

$$B_P = \left(\frac{\mu_0 i}{4r} + \frac{\mu_0 i}{4\pi r}\right) = \frac{\mu_0 i}{2r}\left(\frac{1}{2} + \frac{1}{2\pi}\right)$$

Q21 (4)

$$B_C = \frac{\mu_0 I}{4\pi R}(\pi) \text{ (B at centre of circular arc)}$$

$$= \frac{\mu_0 I}{4R} = \frac{4\pi \times 10^{-7} \times 3}{4 \times \frac{\pi}{10}}$$

$$= 3 \times 10^{-6}\ T = 3\mu T$$

Q22 (2)

$\phi = \vec{B}.\vec{A}$

$= BA \cos\theta$

Most suitable ans is 2 [Otherwise ABCD]

Q23 (44)

$N = 600, A = 70 \times 10^{-4}\ m^2, B = 0.4T$

$$\omega = \frac{500 \times 2\pi}{60} = \frac{100\pi}{6} \text{ rad / s}$$

$E = NAB\omega \sin\omega t$ $\quad\omega t$ is angle b/w $\vec{A}$ & $\vec{B}$

$$= 600 \times 70 \times 10^{-4} \times 0.4 \times \frac{100\pi}{6} \times \frac{1}{2}$$

= 44 V